

Chapter 10

Periodic review base stock policy

A periodic review inventory model places orders to replenish inventory at specific intervals of time. The model is periodic in the sense that ordering decisions are made at fixed time intervals. In contrast to the newsvendor model, inventory that remains at the end of a period can be used in the subsequent period; in the simplest case, inventory does not spoil or deteriorate. The optimization problem again balances the cost of having too much or too little inventory, but the multi-period nature of the problem setting changes the underlying inventory dynamics.

10.0.1 Problem inputs

The *review period*, r , is the interval of time that elapses before an inventory evaluation occurs. For example, if a company is open Monday through Friday then r equal to 5 days denotes a setting where the company evaluates its inventory on a weekly basis.

The *replenishment time*, L , is the elapsed time between placing an order and having the ordered units available to satisfy demand.

$D(a, b)$ is a random variable that denotes the demand over a continuous interval from time a to b . $D(a, b)$ has expectation $(b - a)\mu$ and variance $(b - a)\sigma^2$, where μ and σ^2 are the mean and variance of demand per unit time.

An *order cycle* is the length of time between two consecutive orders while the *replenishment cycle* is the time between two consecutive replenishments, where replenishments are defined as orders received into on-hand inventory). If the replenishment time is deterministic and orders are placed at consecutive review periods the order cycle and the replenishment cycle have the same length. The *cycle service level*, α , represents the probability the SKUL will have sufficient inventory on-hand to meet all demand in an order cycle. The *volume fill rate*, β , represents the fraction of total demand satisfied in an order cycle.

10.0.2 Simplifying assumptions

The following assumptions streamline the derivation of the base stock policy. First, it is assumed that $L < r$. Assuming the replenishment time is less than the review period restricts the number of outstanding orders to zero or one. All the results that follow still hold if $L > r$, but assuming $L < r$ makes the resulting dynamics much easier to represent graphically, and algebraically.

Second, the problem operates in continuous time. This allows us to ignore whether the boundaries a and b in $D(a, b)$ are open or closed. In effect, we assume that the probability there is a demand occurrence at any time instance is negligible and can be ignored. We further assume that demands in non-overlapping time intervals are independent.

We assume any demand that exceeds on-hand inventory is backlogged. Unlike in the newsvendor model, in the periodic review inventory model there will be no lost sales. Instead, unmet demand will be backlogged and satisfied when sufficient replenishments arrive. The resulting derivations implicitly assume inventory levels are chosen such that the system will achieve a high service level, and as such backlogged demand is kept to a minimum.

10.0.3 Graphical depiction of inventory dynamics

A positive replenishment time, L , means any units ordered can not immediately satisfy demand. To reflect this reality, it is worthwhile to distinguish between *inventory on hand* and *inventory position*. Inventory on hand are the units available at the location to satisfy demand. Inventory position is the sum of the inventory on hand and inventory on order.

A base stock policy is a special case of a periodic review inventory model. In a base stock policy, the quantity ordered at each review period equals the amount necessary to raise the inventory position back up to the base stock level. Graphically, the inventory position is illustrated in Figure 10.1:

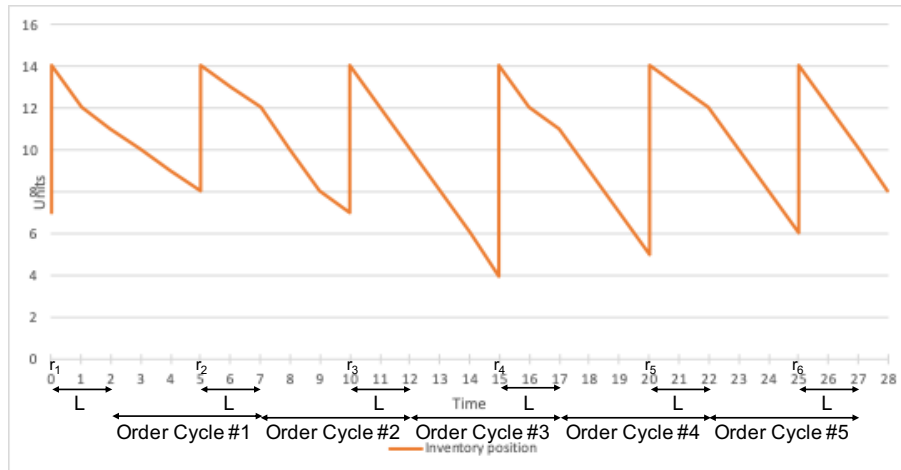


Figure 10.1: This figure illustrates the inventory position for a specific sample path of demand for a SKUL operating a base stock policy. The base stock level is 14 units. Inventory on hand is evaluated at time periods that are an integer multiple of $r = 5$.

In Figure 10.1, the review period, r , is 5 and the base stock level is 14 units. Starting with time $t = 0$, every r periods an order is placed that raises the inventory position back up to base stock level of 14 units. This order causes the jump in the inventory position at every time $t = kr$ where k is a nonnegative integer $0, 1, 2, \dots$

Inventory position counts both inventory on hand and inventory on order. However, only inventory on hand can satisfy demand requests. Figure 10.2 represents the inventory on hand associated with the example in Figure 10.1.

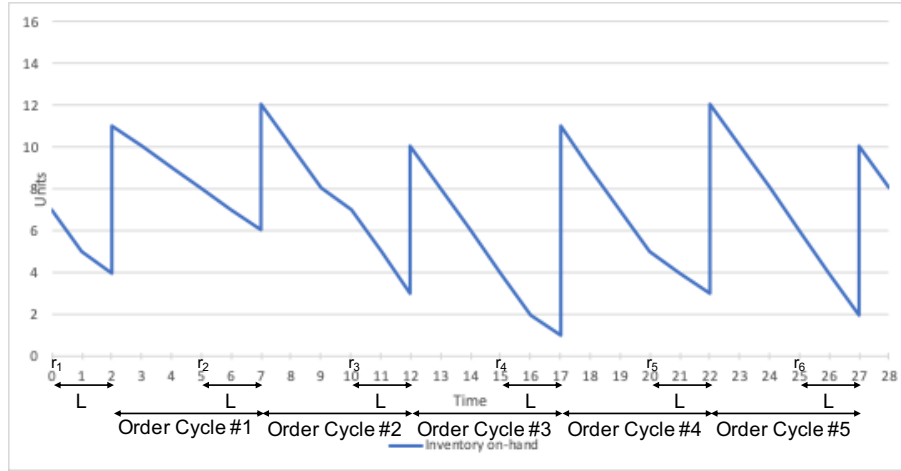


Figure 10.2: This figure illustrates the inventory on hand for a specific sample path of demand for a SKUL operating a base stock policy. The base stock level is 14 units. Inventory on hand is evaluated at time periods that are an integer multiple of $r = 5$ and the amount ordered arrives $L = 2$ periods later.

The base stock level is still 14 units. For this sample path of demand, the maximum inventory on hand is 12 units. Inventory on hand will only equal the base stock level if there is zero demand between an order placed at a time $t = kr$ and that order's arrival at $t = kr + L$.

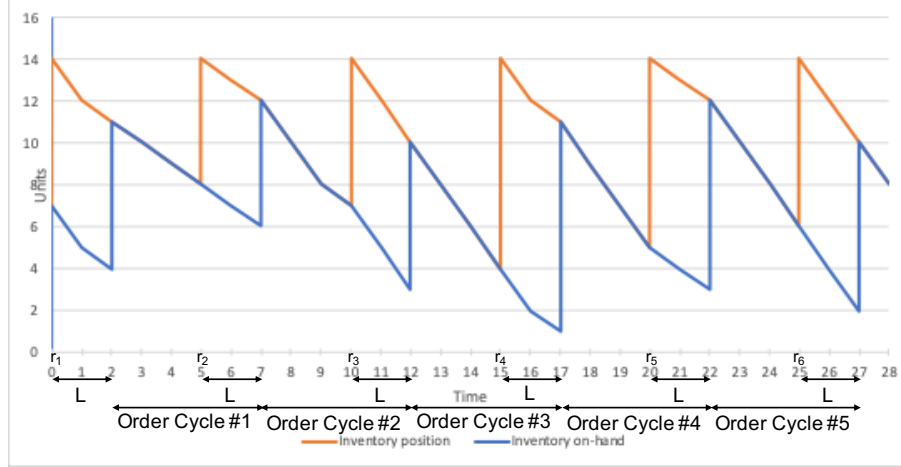


Figure 10.3: This figure illustrates the inventory position and inventory on hand for a specific sample path of demand for a SKUL operating a base stock policy. The base stock level is 14 units. Inventory on hand is evaluated at time periods that are an integer multiple of $r = 5$ and the amount ordered arrives $L = 2$ periods later.

Figure 10.3 combines Figure 10.1 and Figure 10.2 onto one graph. For periods $t \in (kr, kr + L)$ the lines for inventory position and inventory on hand are parallel, and differ by the amount ordered at time $t = kr$. For periods $t \in (kr + L, (k + 1)r)$ there is no order outstanding so the inventory position and inventory on hand are identical.

Figure 10.4 depicts the inventory position and inventory on hand for the same sample path but the base stock level has been changed from 14 units to 12 units. In this case, at time $t = 16$ inventory on hand falls below zero. This negative inventory on hand corresponds to a backorder event where the backordered demand is satisfied at $t = 17$ when the order placed at time $t = 15$ arrives.

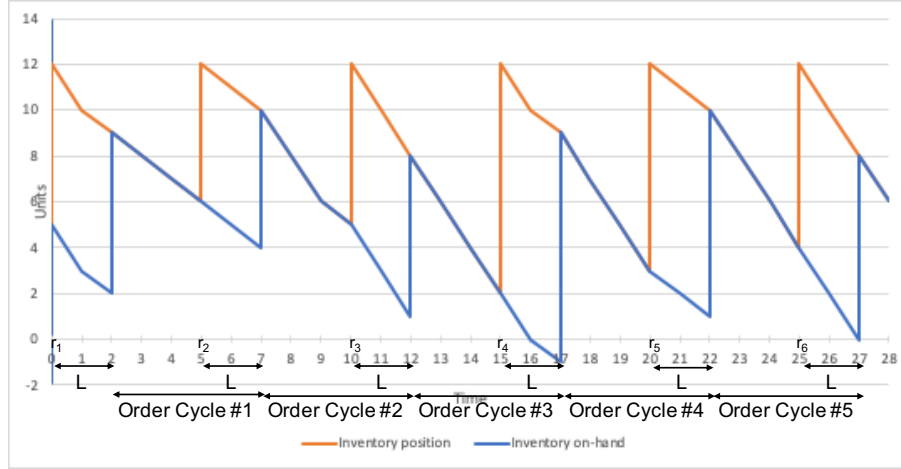


Figure 10.4: This figure illustrates the inventory position and inventory on hand for a specific sample path of demand for a SKUL operating a base stock policy equal to 12 units. A backorder situation begins at time $t = 16$ and is cleared at time $t = 17$.

10.0.4 Algebraic derivation of inventory dynamics

We assume a base stock policy, whereby at each review period we will order up to a base stock level B . This policy is also known as an order-up-to policy, because at every review period the order amount brings the inventory position back up to the base stock level. We will show that this ordering policy operates as a pull policy whereby each order is set to exactly replenish the demand since the last order. That is, at each replenishment epoch, we order an amount equal to the demand since the last replenishment epoch: at time $t = r$, we order $D(0, r)$; at time $t = 2r$, we order $D(r, 2r)$; at time $t = 3r$, we order $D(2r, 3r)$, etc. These orders are received into inventory at times $t = r + L, 2r + L, 3r + L, \dots$

Figure 10.5 shows a representative inventory-time profile for a periodic-review base-stock system. The solid line corresponds to the on-hand inventory. The dotted line is the inventory position, which equals the on-hand inventory plus the on-order inventory.

In Figure 10.5 a replenishment order is placed at the review periods, $t = 0, r, 2r$. At each review period, the replenishment order brings the inventory position up to the base stock level B . The replenishment order is received after the replenishment time of L periods. During this replenishment time the inventory position differs from the on-hand inventory by the order quantity, namely the amount that is on order. Once the replenishment order is received the inventory position equals the on-hand inventory, as there is no longer any on-order inventory.

Let us define $I(t)$ to be a random variable denoting the *inventory on hand* at time t . Let us also define $Q(0)$ as the replenishment order placed at time t . Consider time $t = L$ at which time the order, $Q(0)$ is received. For such epochs, we will define $I(t^+)$ as the on-hand inventory

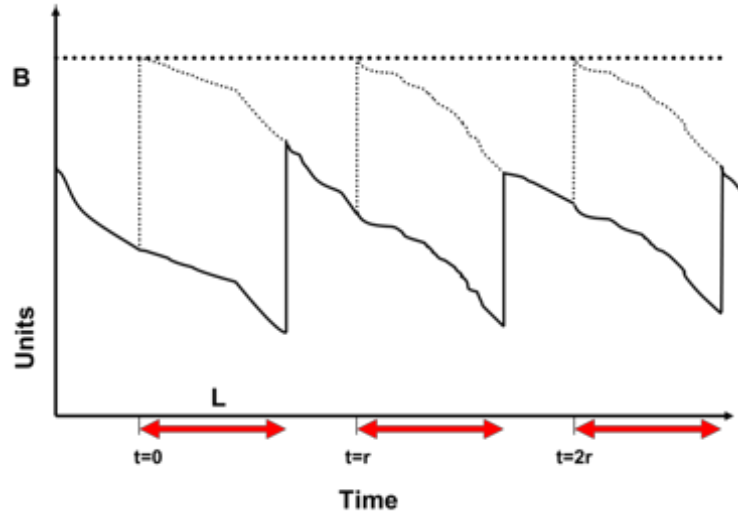


Figure 10.5: On-order and on-hand inventory dynamics for a periodic review, base stock policy.

after receipt of the order, and $I(t^-)$ as the on-hand inventory just prior to the receipt of the order. From Figure 10.5 and the assumptions about the system, we can make a number of observations:

$$I(L^+) = B - D(0, L) \quad (10.1)$$

$$I(r + L^+) = B - D(r, r + L) \quad (10.2)$$

$$I(r + L^-) = B - D(0, r + L) \quad (10.3)$$

As explanation, consider (10.1). At time 0, the order amount $Q(0) = B - I(0)$ brings the inventory position to B . At time L , the order is received, thereby making the on-hand inventory what was on hand at time 0, plus what was ordered, net of the demand from time 0 to time L . As such, $I(L^+)$ is derived as:

$$\begin{aligned} I(L^+) &= I(0) + Q(0) - D(0, L) \\ &= I(0) + (B - I(0)) - D(0, L) \\ &= B - D(0, L) \end{aligned} \quad (10.4)$$

We can make a similar argument for (10.2) since, like $I(L^+)$, $I(r + L^+)$ is also the epoch just after a replenishment order arrives. By definition, $I(r + L^+)$ is the subsequent replenishment cycle after the replenishment cycle ending at time L .

$$\begin{aligned} I(r + L^+) &= I(r) + Q(r) - D(r, r + L) \\ &= I(r) + (B - I(r)) - D(r, r + L) \\ &= B - D(r, r + L) \end{aligned} \quad (10.5)$$

For (10.3), the inventory just before the replenishment at time $t = r + L$ is the inventory after the prior replenishment at time $t = L$, net of the demand from time L to time $r + L$:

$$\begin{aligned} I(r + L^-) &= I(L^+) - D(L, r + L) \\ &= B - D(0, L) - D(L, r + L) \\ &= B - D(0, r + L) \end{aligned} \quad (10.6)$$

Where we use the identity: $D(a, b) + D(b, c) = D(a, c)$. While (10.6) relied on the definition of $I(L^+)$ from (10.1), it is also possible to directly derive (10.3) as follows:

$$\begin{aligned} I(r + L^-) &= I(0) - D(0, L) + Q(0) - D(L, r + L) \\ &= I(0) + Q(0) - D(0, L) - D(L, r + L) \\ &= B - D(0, r + L) \end{aligned} \quad (10.7)$$

Finally, we observe that:

$$\begin{aligned} Q(r) &= D(0, r) \\ Q(2r) &= D(r, 2r) \end{aligned} \quad (10.8)$$

As explanation, we note that by the specification of the base-stock policy we have:

$$\begin{aligned} Q(r) &= B - I(r) \\ Q(2r) &= B - I(2r) \end{aligned} \quad (10.9)$$

From Figure 10.5 and the assumptions that $L < r$ we see that

$$\begin{aligned} I(r) &= B - D(0, r) \\ I(2r) &= B - D(r, 2r) \end{aligned} \quad (10.10)$$

By combining (10.9) and (10.10), we can then obtain (10.8) as follows.

$$\begin{aligned} Q(r) &= B - I(r) & Q(2r) &= B - I(2r) \\ I(r) &= B - D(0, r) & I(2r) &= B - D(r, 2r) \\ Q(r) &= B - (B - D(0, r)) & Q(2r) &= B - (B - D(r, 2r)) \\ Q(r) &= D(0, r) & Q(2r) &= D(r, 2r) \end{aligned} \quad (10.11)$$

In general, for this system, we have the following characterization of the replenishment orders and the on-hand inventory at the end of each replenishment cycle for any positive integer k :

$$\begin{aligned} Q(kr) &= D((k-1)r, kr) \\ I(kr + L^+) &= B - D(kr, kr + L) \\ I(kr + L^-) &= B - D((k-1)r, kr + L) \end{aligned} \quad (10.12)$$

We can then use this characterization to understand the cost of the base stock policy and its service level. In addition, from Figure 10.5 and (10.12) we can also model the on-order inventory, denoted by $P(t)$ for time t , where P stands for pipeline inventory. We see that for the k^{th} review period:

$$\begin{aligned} P(t) &= Q(kr) = D((k-1)r, kr) \text{ for } kr < t \leq kr + L \\ P(t) &= 0 \text{ for } kr + L < t \leq (k+1)r \end{aligned} \quad (10.13)$$

That is, after each review epoch at time $t = kr$, we have on order $Q(kr) = D((k-1)r, kr)$ for the lead time, equal to L . This is pipeline stock, also referred to as pipeline inventory, in the system.

From (10.12) we model the "low point for the on-hand inventory in a replenishment cycle, as well as the "high" point. We can determine an expectation for each of these as:

$$\begin{aligned} E[I(kr + L^+)] &= B - \mu L \\ E[I(kr + L^-)] &= B - \mu(r + L) \end{aligned} \quad (10.14)$$

We approximate the *average inventory on hand* for this system by averaging these two terms:

$$E[I] = B - \mu \left(\frac{r}{2} + L \right) \quad (10.15)$$

This is an approximation of the average on-hand inventory, mainly because it counts backorders as negative inventory. A more accurate model would count the inventory as being zero, when there are backorders.

Equation (10.13) specifies the pipeline inventory for a review period, namely from $t = kr$ to $t = (k+1)r$. From $t = kr$ to $t = kr + L$ there is a pipeline inventory equal to $Q(kr) = D((k-1)r, kr)$. From $t = kr + L$ to $t = (k+1)r$ there is zero pipeline inventory. To determine the average pipeline inventory over the review period, we take a time-weighted average over the review period:

$$\begin{aligned} E[P] &= \left(\frac{L}{r} \right) (E[D((k-1)r, kr)]) + \left(\frac{r-L}{r} \right) (0) \\ &= \left(\frac{L}{r} \right) (\mu r) \\ &= \mu L \end{aligned} \quad (10.16)$$

We see that the average pipeline inventory does not depend on the review period, but is just the expected demand over the lead time. This finding is quite general and will remain true for many model extensions and variations.

By considering the low point for the inventory we can also develop expressions for the service levels. The *cycle service level* (CSL) is the probability of no shortages or backorders during a replenishment event; it is found to be:

$$\begin{aligned}
CSL &= \Pr[I(kr + L^-) \geq 0] \\
&= \Pr[B - D((k-1)r, kr + L) \geq 0] \\
&= \Pr[D((k-1)r, kr + L) \leq B]
\end{aligned} \tag{10.17}$$

Thus, the CSL is the probability that demand over a time interval of length $r + L$ is less than the base stock B . We see here that in order to provide a high service level, we need to set the base stock B to cover the demand over the review period plus replenishment lead time, namely over a time interval of length $r + L$. For this reason we define the review period plus replenishment lead time to be the *coverage time* or *coverage interval*.

We note that demand over the coverage time has mean equal to $(r + L)\mu$, and a variance equal to $(r + L)\sigma^2$.

If we assume that demand is normally distributed, then we find that $CSL = \Phi(z)$ where the safety factor z is given as $z = \frac{B - (r+L)\mu}{\sigma\sqrt{r+L}}$ and $\Phi()$ is the cumulative distribution function for the standard normal random variable.

Sometimes a service level target is set. For instance, we might be given a service target $CSL = \alpha$ for some value. Then we need to set the base stock as:

$$\begin{aligned}
B &= (r + L)\mu + z\sigma\sqrt{r + L} \\
\text{where } \Phi(z) &= \alpha
\end{aligned} \tag{10.18}$$

For instance, if $\alpha = 0.977$ then we set the safety factor $z = 2$, and set the base stock as shown in equation (10.18).

The net result of all this reading is that under a base stock policy, the average inventory on-hand is equal to $\frac{r\mu}{2} + z\sigma\sqrt{r + L}$ where the first term, $\frac{r\mu}{2}$, defines the expected cycle stock and the second term, $z\sigma\sqrt{r + L}$, defines the expected safety stock. Furthermore, the expected inventory on-order is $L\mu$ which is defined as the expected pipeline stock.

Graphically, this is illustrated in Figure 10.6:

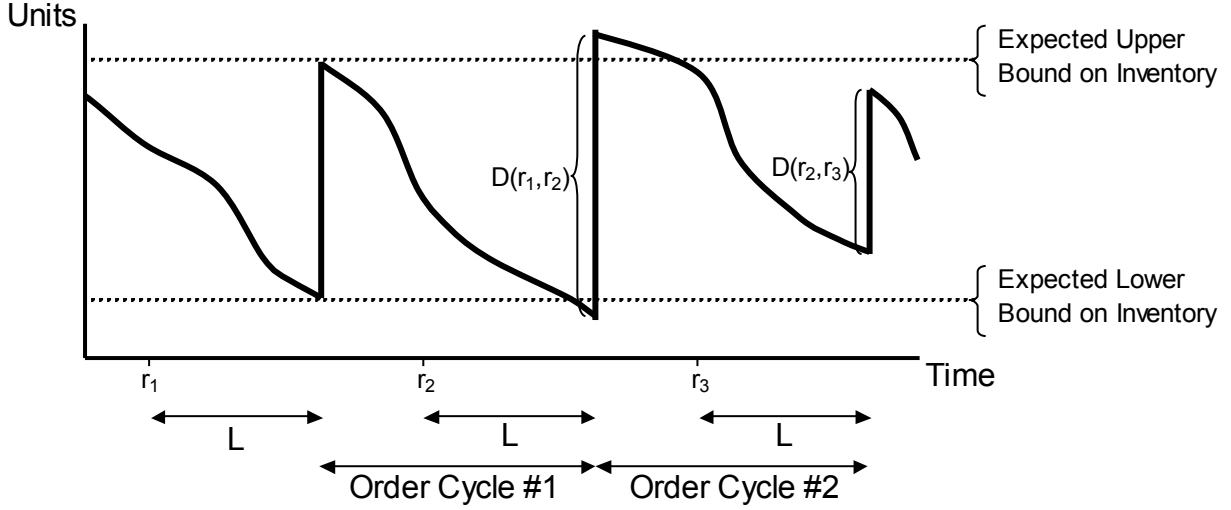


Figure 10.6: This figure illustrates the inventory on hand for a specific sample path of demand for a SKUL that operates a base stock policy. Inventory is evaluated at time periods that are an integer multiple of r and the amount ordered arrives L periods later.

ordered since the last review period. One can think of a base stock policy as a discrete time kanban system where the amount ordered equals the total amount consumed since the last order.

If the review period length is r , then reviews occur at integer multiples of r . If we call the i -th review period r_i then the order placed at period r_i is the total demand consumed between r_{i-1} and r_i . This is represented as $D(r_{i-1}, r_i)$ and it is a random variable that aggregates r periods of demand.

But characterizing $D(r_{i-1}, r_i)$ is not enough because we do not assume orders can be satisfied immediately. Instead, we assume there is a leadtime of L periods between the placing of an order and its arrival into on-hand inventory. To characterize the average inventory for this base stock system, we first characterize the average inventory for one order cycle. We define an order cycle as the time between two consecutive arrivals of inventory. An order cycle could instead be defined as the time between two successive review periods (because the interval of time is the same) but the inventory dynamics between two successive order arrivals is easier to characterize mathematically. Once we can characterize the inventory over an order cycle, we can recognize that in expectation all order cycles will behave identically because demand is i.i.d. and we are maintaining the same base stock policy across all periods.

Whereas the Figure 10.6 was a sample path that showed a specific realization of demands, the Figure 10.7 represents the system's expected behavior:

Graphically, we can think of the inventory on hand in an order cycle as a triangle on top of a rectangle. The triangle is the cycle stock. At its maximum it equals $r\mu$ where μ is average demand per base time period (i.e., the base time period is one business day) and r is the number of base time periods between reviews (i.e., r equal to five denotes a weekly review period).

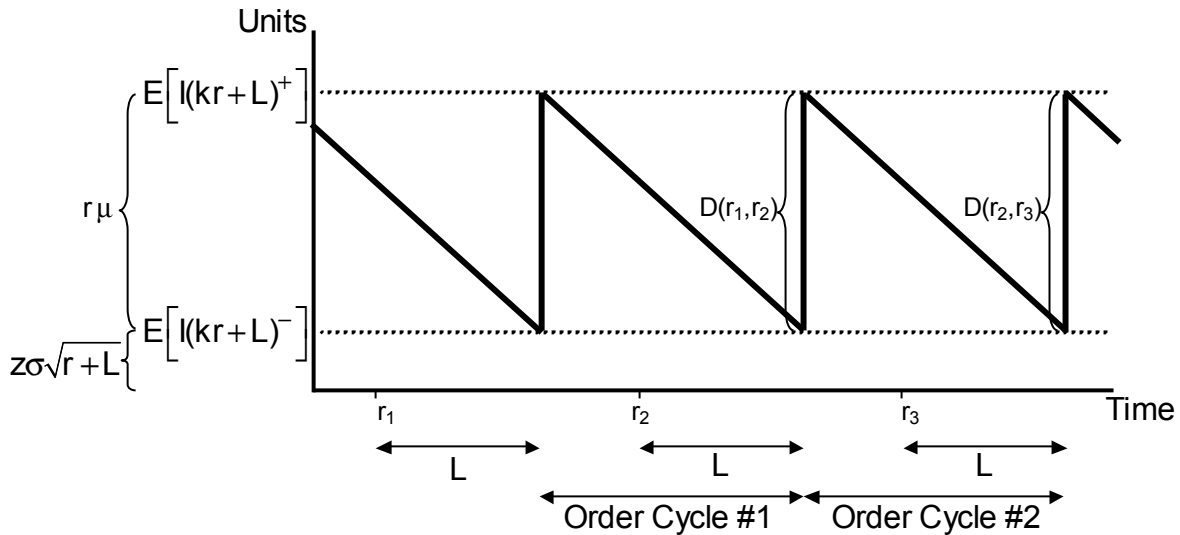


Figure 10.7: This figure illustrates the expected inventory on hand for a SKUL that operates a base stock policy. Inventory is evaluated at time periods that are an integer multiple of r and the amount ordered arrives L periods later.

The height of the rectangle is equal to $z\sigma\sqrt{r+L}$ where σ is the standard deviation of demand in a base time period. $r+L$ is referred to as the replenishment leadtime. It is the maximum amount of time that is exposed to a potential stockout before replenishment occurs. Informally speaking, think of a huge demand that occurs right after a review period. That demand spike will not even be recognized until the following review period and the resulting order will not be filled for another L periods. Now over the course of those $r+L$ periods some days will experience high demands and some periods will experience low demands. We need to characterize the standard deviation of demand over the entire replenishment leadtime and then protect against the variability in demand over that interval of time. If the demand per base time period is $N(\mu_i, \sigma_i)$ and $z = \text{NORMSINV}(\alpha)$ where α is the service level, then this is precisely what $z\sigma\sqrt{r+L}$ is doing.

Under a base stock policy, the average inventory is equal to $\frac{r\mu}{2} + z\sigma\sqrt{r+L}$

The first term, $\frac{r\mu}{2}$, defines the expected cycle stock and the second term, $z\sigma\sqrt{r+L}$, defines the expected safety stock.

Cycle and safety stock are completed units held in finished goods inventory after the processing activity (which takes L periods) is completed. Along with these finished goods, there is also work in process inventory. We refer to this work in process as pipeline stock inventory. The expected pipeline stock inventory equals $L\mu$ units.

10.0.5 Contrasting newsvendor and base stock inventory models

Since the newsvendor model is a single-period model, in the newsvendor model it is sufficient to assume the order is placed before demand materializes and that the supply arrives before demand materializes; the actual replenishment time for an order in the newsvendor model is an operational detail that can be ignored. In contrast, the multi-period nature of the periodic review inventory model requires a proper accounting of the replenishment process.

10.0.6 Modeling discrete time versus continuous time

Up to this point in Chapter 10, time has been continuous. Continuous time is useful to unclutter the base stock derivation. However, most problems in practice are best modeled in discrete time. The underlying ideas presented do not change, but discrete time does introduce some more complexity that needs to be taken into account.